

Fossil - tablet error

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```
library(tibble)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(ggplot2)
```

1 Baseline FOVS simualtion

```
simulate_fovs_once <- function(
  n_target = 10000,
  N1 = 1,
  tablet_mean = 5000,
  tablet_cv = 0,
  region_side = 1,
  fov_size = 0.1,
  n_cal = 20,
  n_ext = 20,
  omega = 2
) {

  # Sample area
  V <- region_side^2

  # True concentration
  true_conc <- n_target / V

  # Generate target particles
```

```

points <- tibble(
  x = runif(n_target, min = 0, max = region_side),
  y = runif(n_target, min = 0, max = region_side)
)

# Known average marker dose used in the estimator
known_marker_dose <- N1 * tablet_mean

# Generate per-tablet marker doses, then sum them
tablet_sd <- tablet_mean * tablet_cv
tablet_doses <- rnorm(N1, mean = tablet_mean, sd = tablet_sd)
tablet_doses <- pmax(round(tablet_doses), 1)

true_marker_total <- sum(tablet_doses)

# Generate marker particles
markers <- tibble(
  x = runif(true_marker_total, min = 0, max = region_side),
  y = runif(true_marker_total, min = 0, max = region_side)
)

# Build grid of candidate FOV locations
n_side <- region_side / fov_size
if (abs(n_side - round(n_side)) > 1e-10) {
  stop("fov_size must divide the region side exactly.")
}

all_grid <- expand_grid(
  x = seq(0, region_side - fov_size, by = fov_size),
  y = seq(0, region_side - fov_size, by = fov_size)
)

# Remove outermost edge cells
inner_grid <- subset(
  all_grid,
  x > 0 & x < (region_side - fov_size) &
  y > 0 & y < (region_side - fov_size)
)

if (nrow(inner_grid) < n_cal + n_ext) {
  stop("Not enough inner grid cells for the requested number of FOVs.")
}

# Define a shared local window so calibration and extrapolation
# FOVs come from nearby locations
window_cols <- 7
window_rows <- 6

x_vals <- sort(unique(inner_grid$x))
y_vals <- sort(unique(inner_grid$y))

inner_nx <- length(x_vals)
inner_ny <- length(y_vals)

```

```

if (window_cols > inner_nx || window_rows > inner_ny) {
  stop("Shared FOV window is too large for the inner grid.")
}

start_col <- sample(1:(inner_nx - window_cols + 1), size = 1)
start_row <- sample(1:(inner_ny - window_rows + 1), size = 1)

window_x <- x_vals[start_col:(start_col + window_cols - 1)]
window_y <- y_vals[start_row:(start_row + window_rows - 1)]

shared_grid <- subset(
  inner_grid,
  x %in% window_x & y %in% window_y
)

if (nrow(shared_grid) < n_cal + n_ext) {
  stop("Not enough cells in the shared local FOV region.")
}

# Select calibration and extrapolation FOVs from the same local region
cal_idx <- sample(1:nrow(shared_grid), size = n_cal, replace = FALSE)
cal_fovs <- shared_grid[cal_idx, ]

remaining_grid <- shared_grid[-cal_idx, ]
ext_idx <- sample(1:nrow(remaining_grid), size = n_ext, replace = FALSE)
ext_fovs <- remaining_grid[ext_idx, ]

# Count fossils in calibration FOVs
cal_fossil <- numeric(n_cal)

for (i in 1:n_cal) {
  cal_fossil[i] <- sum(
    points$x >= cal_fovs$x[i] &
    points$x < cal_fovs$x[i] + fov_size &
    points$y >= cal_fovs$y[i] &
    points$y < cal_fovs$y[i] + fov_size
  )
}

# Count markers in extrapolation FOVs
ext_marker <- numeric(n_ext)

for (i in 1:n_ext) {
  ext_marker[i] <- sum(
    markers$x >= ext_fovs$x[i] &
    markers$x < ext_fovs$x[i] + fov_size &
    markers$y >= ext_fovs$y[i] &
    markers$y < ext_fovs$y[i] + fov_size
  )
}

# Calculate effort (time proxy)
x_total <- sum(cal_fossil)

```

```

n_count <- sum(ext_marker)
effort <- (omega * n_cal) + x_total + (omega * n_ext) + n_count

# Calculate concentration and error
ybar <- mean(cal_fossil)
xhat <- ybar * n_ext

failed <- (n_count == 0 || ybar == 0)

if (failed) {
  conc <- NA
  tablet_term <- NA
  calibration_term <- NA
  marker_term <- NA
  total_error <- NA
  bias <- NA
  percent_bias <- NA
  s3p <- NA
} else {
  conc <- (xhat * known_marker_dose) / (n_count * V)

  s3p <- sd(cal_fossil) / mean(cal_fossil)

  # Error decomposition
  tablet_term <- 100 * (tablet_cv / sqrt(N1))
  calibration_term <- 100 * (s3p / sqrt(n_cal))
  marker_term <- 100 * (sqrt(n_count) / n_count)

  total_error <- sqrt(
    tablet_term^2 +
    calibration_term^2 +
    marker_term^2
  )

  bias <- conc - true_conc
  percent_bias <- 100 * bias / true_conc
}

tibble(
  true_conc = true_conc,
  known_marker_dose = known_marker_dose,
  true_marker_total = true_marker_total,
  counted_markers = n_count,
  ybar = ybar,
  s3p = s3p,
  conc = conc,
  tablet_term = tablet_term,
  calibration_term = calibration_term,
  marker_term = marker_term,
  total_error = total_error,
  bias = bias,
  percent_bias = percent_bias,
  effort = effort,

```

```

    failed = failed
  )
}

```

2 Repeated simulations

```

simulate_fovs_repeated <- function(
  n_rep = 5000,
  n_target = 10000,
  N1 = 1,
  tablet_mean = 5000,
  tablet_cv = 0,
  region_side = 1,
  fov_size = 0.1,
  n_cal = 20,
  n_ext = 20,
  omega = 2
) {

  results <- vector("list", n_rep)

  for (k in 1:n_rep) {
    results[[k]] <- simulate_fovs_once(
      n_target = n_target,
      N1 = N1,
      tablet_mean = tablet_mean,
      tablet_cv = tablet_cv,
      region_side = region_side,
      fov_size = fov_size,
      n_cal = n_cal,
      n_ext = n_ext,
      omega = omega
    )
  }

  results_df <- bind_rows(results)

  mean_conc <- mean(results_df$conc, na.rm = TRUE)
  sd_conc <- sd(results_df$conc, na.rm = TRUE)
  mean_error <- mean(results_df$total_error, na.rm = TRUE)
  sim_cv <- 100 * sd_conc / mean_conc
  rmse <- sqrt(mean((results_df$conc - results_df$true_conc)^2, na.rm = TRUE))
  failure_rate <- mean(results_df$failed)

  summary_df <- tibble(
    N1 = N1,
    tablet_cv = tablet_cv,
    mean_true_conc = mean(results_df$true_conc, na.rm = TRUE),
    mean_estimated_conc = mean_conc,
    empirical_sd = sd_conc,

```

```

    empirical_cv = sim_cv,
    mean_tablet_term = mean(results_df$tablet_term, na.rm = TRUE),
    mean_calibration_term = mean(results_df$calibration_term, na.rm = TRUE),
    mean_marker_term = mean(results_df$marker_term, na.rm = TRUE),
    mean_theoretical_error = mean_error,
    mean_bias = mean(results_df$bias, na.rm = TRUE),
    mean_percent_bias = mean(results_df$percent_bias, na.rm = TRUE),
    rmse = rmse,
    mean_effort = mean(results_df$effort, na.rm = TRUE),
    failure_rate = failure_rate
  )

  list(
    raw = results_df,
    summary = summary_df
  )
}

```

Baseline result

```

set.seed(456)

baseline_result <- simulate_fovs_repeated(
  n_rep = 5000,
  n_target = 10000,
  N1 = 1,
  tablet_mean = 5000,
  tablet_cv = 0,
  region_side = 1,
  fov_size = 0.1,
  n_cal = 20,
  n_ext = 20,
  omega = 2
)

baseline_summary <- baseline_result$summary
print(baseline_summary, width = Inf)

## # A tibble: 1 x 15
##       N1 tablet_cv mean_true_conc mean_estimated_conc empirical_sd empirical_cv
##   <dbl>   <dbl>       <dbl>           <dbl>         <dbl>      <dbl>
## 1     1     0     10000         10011.         348.       3.47
##   mean_tablet_term mean_calibration_term mean_marker_term mean_theoretical_error
##             <dbl>             <dbl>             <dbl>             <dbl>
## 1             0             2.20             3.16             3.86
##   mean_bias mean_percent_bias  rmse mean_effort failure_rate
##       <dbl>       <dbl> <dbl>       <dbl>       <dbl>
## 1    11.1         0.111 348.       3079.         0

```

```
baseline_summary %>%
  select(
    tablet_cv,
    mean_true_conc,
    mean_estimated_conc,
    mean_theoretical_error,
    empirical_cv,
    mean_bias,
    rmse,
    mean_effort,
    failure_rate
  ) %>%
  knitr::kable(digits = 2, caption = "Baseline simulation results")
```

Table 1: Baseline simulation results

tablet_cv	mean_true_conc	mean_estimated_conc	mean_theoretical_error	empirical_cv	mean_bias	rmse	mean_effort	failure_rate
0	10000	10011.09	3.86	3.47	11.09	347.83	3079.19	0

Tablet CV scenarios

```
set.seed(456)

tablet_cv_levels <- c(0, 0.01, 0.02, 0.04, 0.06)

results_cv <- vector("list", length(tablet_cv_levels))

for (i in seq_along(tablet_cv_levels)) {
  results_cv[[i]] <- simulate_fovs_repeated(
    n_rep = 5000,
    n_target = 10000,
    N1 = 1,
    tablet_mean = 5000,
    tablet_cv = tablet_cv_levels[i],
    region_side = 1,
    fov_size = 0.1,
    n_cal = 20,
    n_ext = 20,
    omega = 2
  )$summary
}

summary_cv <- bind_rows(results_cv)
print(summary_cv, width = Inf)
```

```
## # A tibble: 5 x 15
##       N1 tablet_cv mean_true_conc mean_estimated_conc empirical_sd empirical_cv
##   <dbl>   <dbl>         <dbl>         <dbl>         <dbl>         <dbl>
## 1     1     0         10000         10011.         348.         3.47
```

```
## 2      1      0.01      10000      10004.      364.      3.64
## 3      1      0.02      10000      10021.      401.      4.00
## 4      1      0.04      10000      10009.      536.      5.35
## 5      1      0.06      10000      10037.      711.      7.08
##      mean_tablet_term mean_calibration_term mean_marker_term mean_theoretical_error
##              <dbl>              <dbl>              <dbl>              <dbl>
## 1              0              2.20              3.16              3.86
## 2              1              2.21              3.16              4.00
## 3              2              2.21              3.17              4.36
## 4              4              2.21              3.16              5.57
## 5              6              2.21              3.17              7.14
##      mean_bias mean_percent_bias rmse mean_effort failure_rate
##              <dbl>              <dbl> <dbl>              <dbl>              <dbl>
## 1      11.1              0.111 348.              3079.              0
## 2       3.99              0.0399 364.              3080.              0
## 3      20.9              0.209 401.              3079.              0
## 4       9.26              0.0926 536.              3081.              0
## 37.1              0.371 712.              3081.              0
```

```
summary_cv %>%
  select(
    tablet_cv,
    mean_true_conc,
    mean_estimated_conc,
    mean_theoretical_error,
    empirical_cv,
    mean_bias,
    rmse,
    mean_effort,
    failure_rate
  ) %>%
  knitr::kable(digits = 2, caption = "Simulation results across tablet CV levels")
```

Table 2: Simulation results across tablet CV levels

tablet_cv	mean_true_conc	mean_estimated_conc	mean_theoretical_error	empirical_cv	mean_bias	rmse	mean_effort	failure_rate
0.00	10000	10011.09	3.86	3.47	11.09	347.83	3079.19	0
0.01	10000	10003.99	4.00	3.64	3.99	364.34	3079.84	0
0.02	10000	10020.93	4.36	4.00	20.93	401.33	3078.95	0
0.04	10000	10009.26	5.57	5.35	9.26	535.87	3081.10	0
0.06	10000	10037.06	7.14	7.08	37.06	711.54	3080.86	0

```
summary_cv <- summary_cv %>%
  mutate(
    tablet_share = mean_tablet_term^2 / mean_theoretical_error^2,
    calibration_share = mean_calibration_term^2 / mean_theoretical_error^2,
    marker_share = mean_marker_term^2 / mean_theoretical_error^2
  )
```

```
summary_cv %>%
  select(
    tablet_cv,
```



```

mean_tablet_term,
mean_calibration_term,
mean_marker_term,
mean_theoretical_error
) %>%
knitr::kable(digits = 2, caption = "Error decomposition across tablet CV levels")

```

Table 3: Error decomposition across tablet CV levels

tablet_cv	mean_tablet_term	mean_calibration_term	mean_marker_term	mean_theoretical_error
0.00	0	2.20	3.16	3.86
0.01	1	2.21	3.16	4.00
0.02	2	2.21	3.17	4.36
0.04	4	2.21	3.16	5.57
0.06	6	2.21	3.17	7.14

```

summary_cv %>%
  select(
    tablet_cv,
    tablet_share,
    calibration_share,
    marker_share
  ) %>%
knitr::kable(digits = 3, caption = "Relative contributions of error components across tablet CV levels")

```

Table 4: Relative contributions of error components across tablet CV levels

tablet_cv	tablet_share	calibration_share	marker_share
0.00	0.000	0.323	0.670
0.01	0.063	0.306	0.626
0.02	0.211	0.257	0.528
0.04	0.516	0.158	0.322
0.06	0.705	0.096	0.196

```

summary_cv_share_long <- summary_cv %>%
  select(
    tablet_cv,
    tablet_share,
    calibration_share,
    marker_share
  ) %>%
  tidyr::pivot_longer(
    cols = c(tablet_share, calibration_share, marker_share),
    names_to = "component",
    values_to = "share"
  )

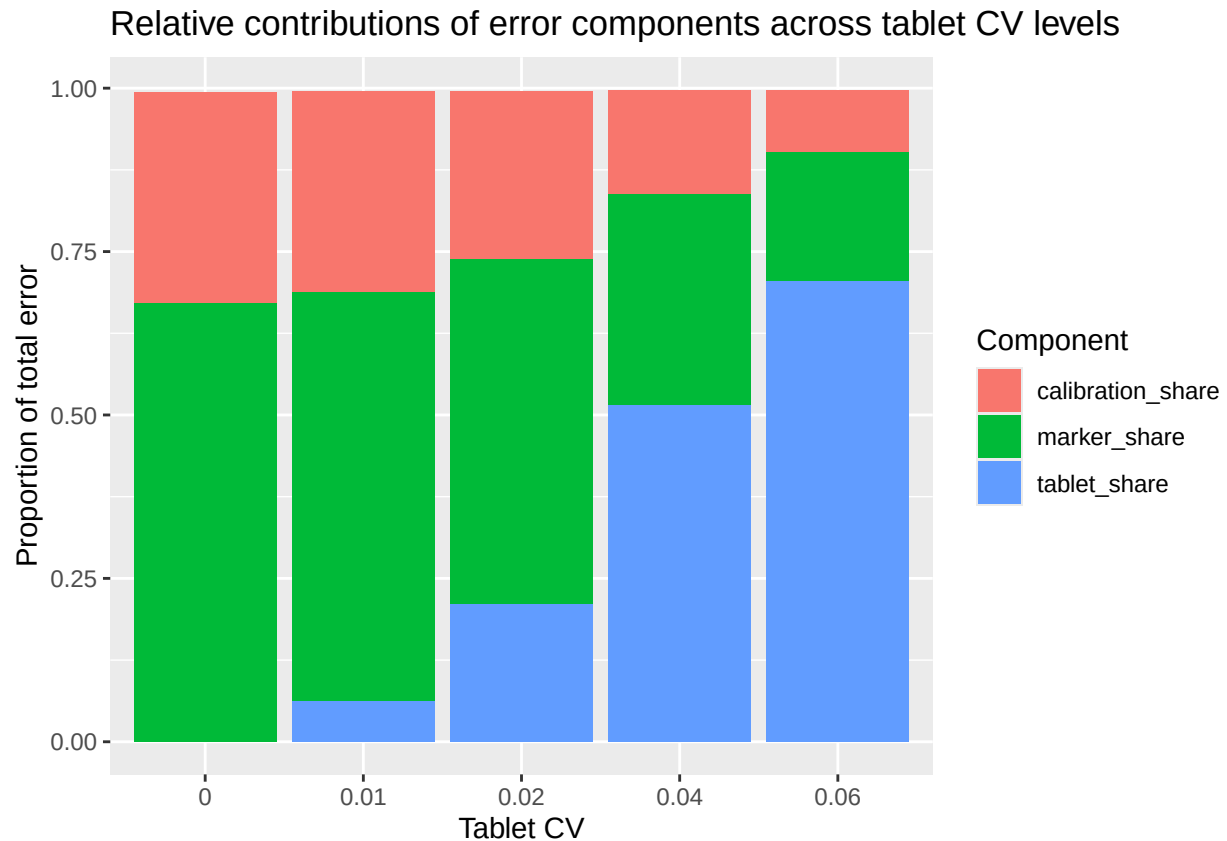
ggplot(summary_cv_share_long, aes(x = factor(tablet_cv), y = share, fill = component)) +
  geom_col() +

```

```

labs(
  title = "Relative contributions of error components across tablet CV levels",
  x = "Tablet CV",
  y = "Proportion of total error",
  fill = "Component"
)

```



N1 scenarios

```

set.seed(456)

N1_levels <- c(1, 2, 4)

results_N1 <- vector("list", length(N1_levels))

for (i in seq_along(N1_levels)) {
  results_N1[[i]] <- simulate_fovs_repeated(
    n_rep = 5000,
    n_target = 10000,
    N1 = N1_levels[i],
    tablet_mean = 5000,
    tablet_cv = 0.04,

```

```

    region_side = 1,
    fov_size = 0.1,
    n_cal = 20,
    n_ext = 20,
    omega = 2
  )$summary
}

summary_N1 <- bind_rows(results_N1)
print(summary_N1, width = Inf)

```

```

## # A tibble: 3 x 15
##       N1 tablet_cv mean_true_conc mean_estimated_conc empirical_sd empirical_cv
##   <dbl>   <dbl>         <dbl>           <dbl>         <dbl>      <dbl>
## 1     1     0.04         10000          10033.         532.        5.30
## 2     2     0.04         10000          10010.         401.        4.01
## 3     4     0.04         10000          10001.         317.        3.17
##   mean_tablet_term mean_calibration_term mean_marker_term mean_theoretical_error
##         <dbl>           <dbl>           <dbl>           <dbl>
## 1             4             2.20             3.17             5.57
## 2            2.83             2.20             2.24             4.24
## 3             2             2.20             1.58             3.38
##   mean_bias mean_percent_bias rmse mean_effort failure_rate
##         <dbl>           <dbl> <dbl>         <dbl>         <dbl>
## 1    33.0           0.330  533.         3078.           0
## 2    10.4           0.104  401.         4080.           0
## 3     0.600         0.00600  317.         6079.           0

```

```

summary_N1 %>%
  select(
    N1,
    tablet_cv,
    mean_true_conc,
    mean_estimated_conc,
    mean_theoretical_error,
    empirical_cv,
    mean_bias,
    rmse,
    mean_effort,
    failure_rate
  ) %>%
  knitr::kable(digits = 2, caption = "Simulation results across numbers of tablets")

```

Table 5: Simulation results across numbers of tablets

N1	tablet_cv	mean_true_conc	mean_estimated_conc	mean_theoretical_error	empirical_cv	mean_bias	rmse	mean_effort	failure_rate
1	0.04	10000	10032.98	5.57	5.30	32.98	532.90	3078.21	0
2	0.04	10000	10010.41	4.24	4.01	10.41	401.16	4080.09	0
4	0.04	10000	10000.60	3.38	3.17	0.60	316.79	6078.58	0

```
summary_N1 <- summary_N1 %>%
  mutate(
    tablet_share = mean_tablet_term^2 / mean_theoretical_error^2,
    calibration_share = mean_calibration_term^2 / mean_theoretical_error^2,
    marker_share = mean_marker_term^2 / mean_theoretical_error^2
  )

summary_N1 %>%
  select(
    N1,
    mean_tablet_term,
    mean_calibration_term,
    mean_marker_term,
    mean_theoretical_error
  ) %>%
  knitr::kable(digits = 2, caption = "Error decomposition across numbers of tablets")
```

Table 6: Error decomposition across numbers of tablets

N1	mean_tablet_term	mean_calibration_term	mean_marker_term	mean_theoretical_error
1	4.00	2.2	3.17	5.57
2	2.83	2.2	2.24	4.24
4	2.00	2.2	1.58	3.38

```
summary_N1 %>%
  select(
    N1,
    tablet_share,
    calibration_share,
    marker_share
  ) %>%
  knitr::kable(digits = 3, caption = "Relative contributions of error components across numbers of tablets")
```

Table 7: Relative contributions of error components across numbers of tablets

N1	tablet_share	calibration_share	marker_share
1	0.516	0.156	0.324
2	0.446	0.270	0.279
4	0.350	0.424	0.219

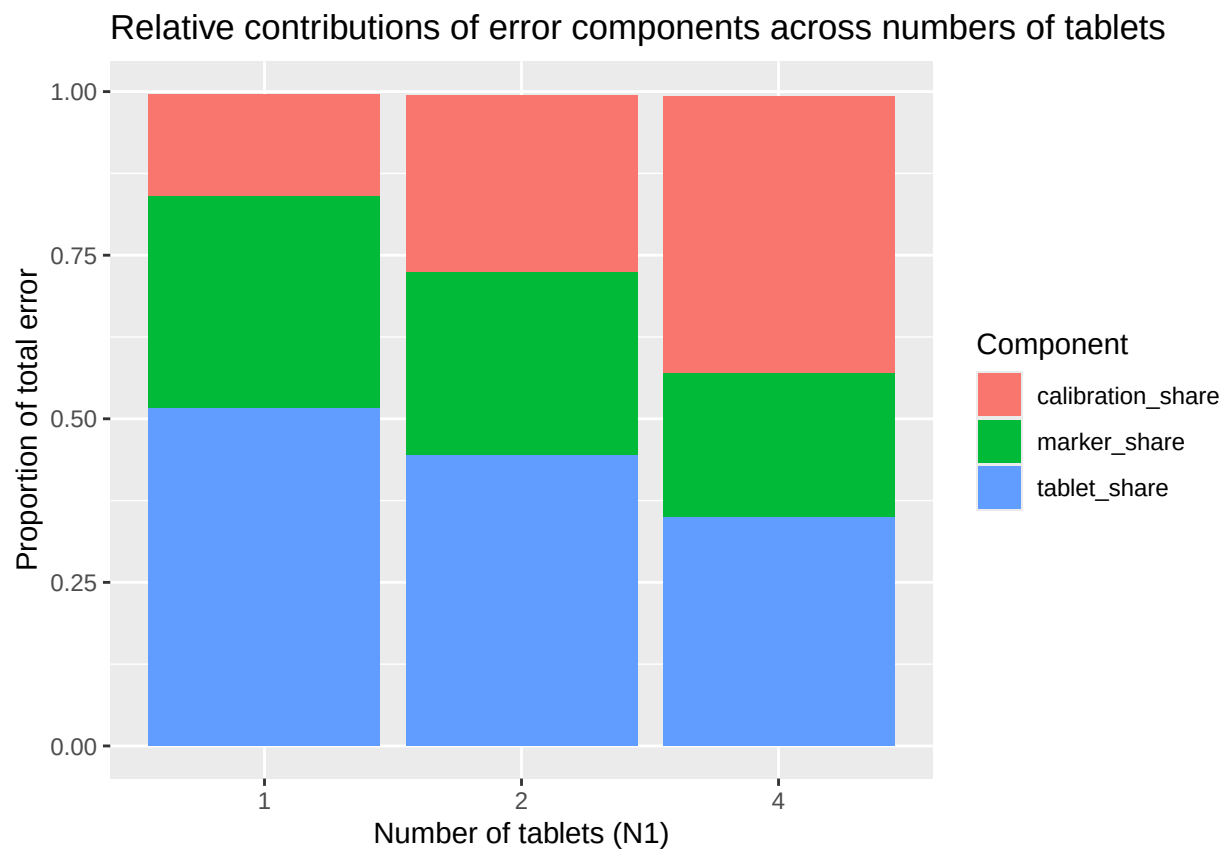
```
summary_N1_share_long <- summary_N1 %>%
  select(
    N1,
    tablet_share,
    calibration_share,
    marker_share
  ) %>%
  tidyr::pivot_longer(
```

```

cols = c(tablet_share, calibration_share, marker_share),
names_to = "component",
values_to = "share"
)

ggplot(summary_N1_share_long, aes(x = factor(N1), y = share, fill = component)) +
  geom_col() +
  labs(
    title = "Relative contributions of error components across numbers of tablets",
    x = "Number of tablets (N1)",
    y = "Proportion of total error",
    fill = "Component"
  )

```



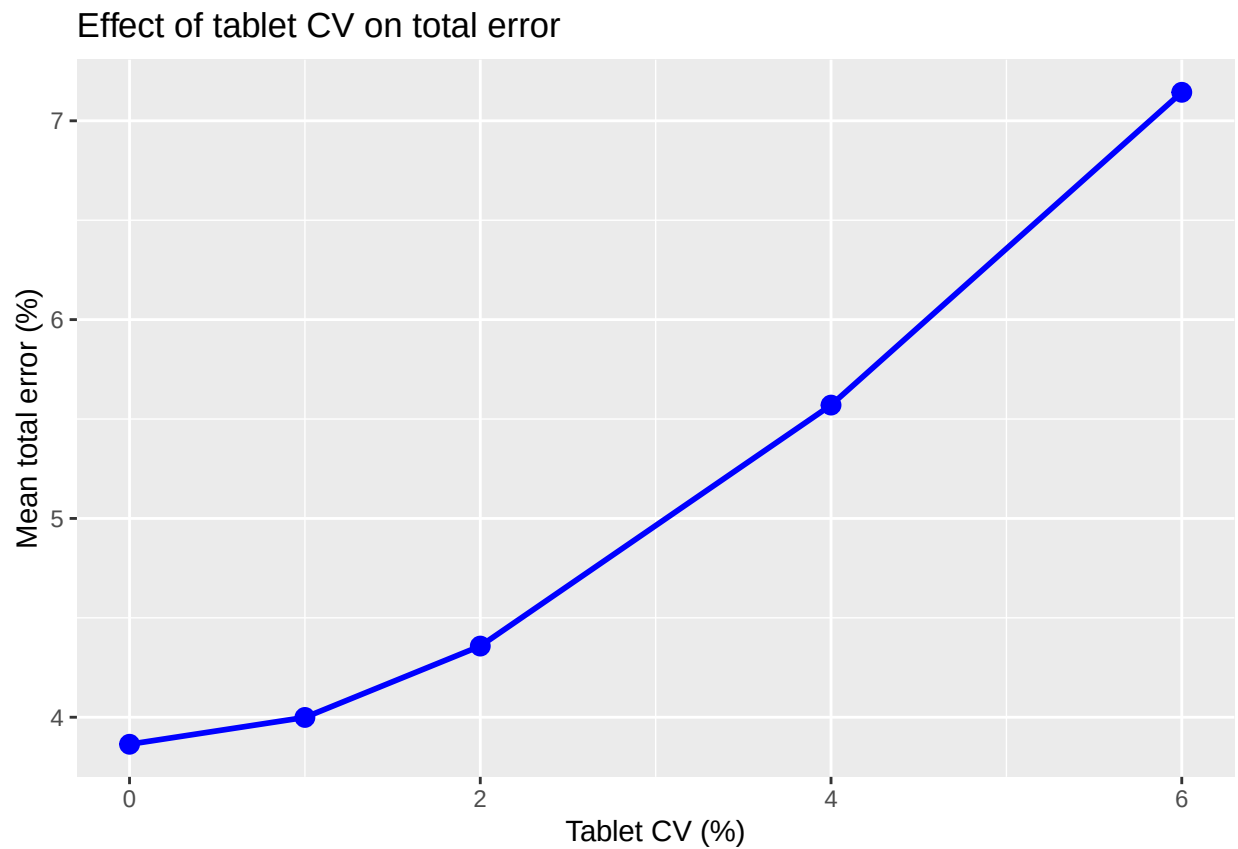
Plot 1: tablet CV and total error

```

ggplot(summary_cv, aes(x = tablet_cv * 100, y = mean_theoretical_error)) +
  geom_line(linewidth = 1, colour = "blue") +
  geom_point(size = 3, colour = "blue") +
  labs(
    title = "Effect of tablet CV on total error",
    x = "Tablet CV (%)",
  )

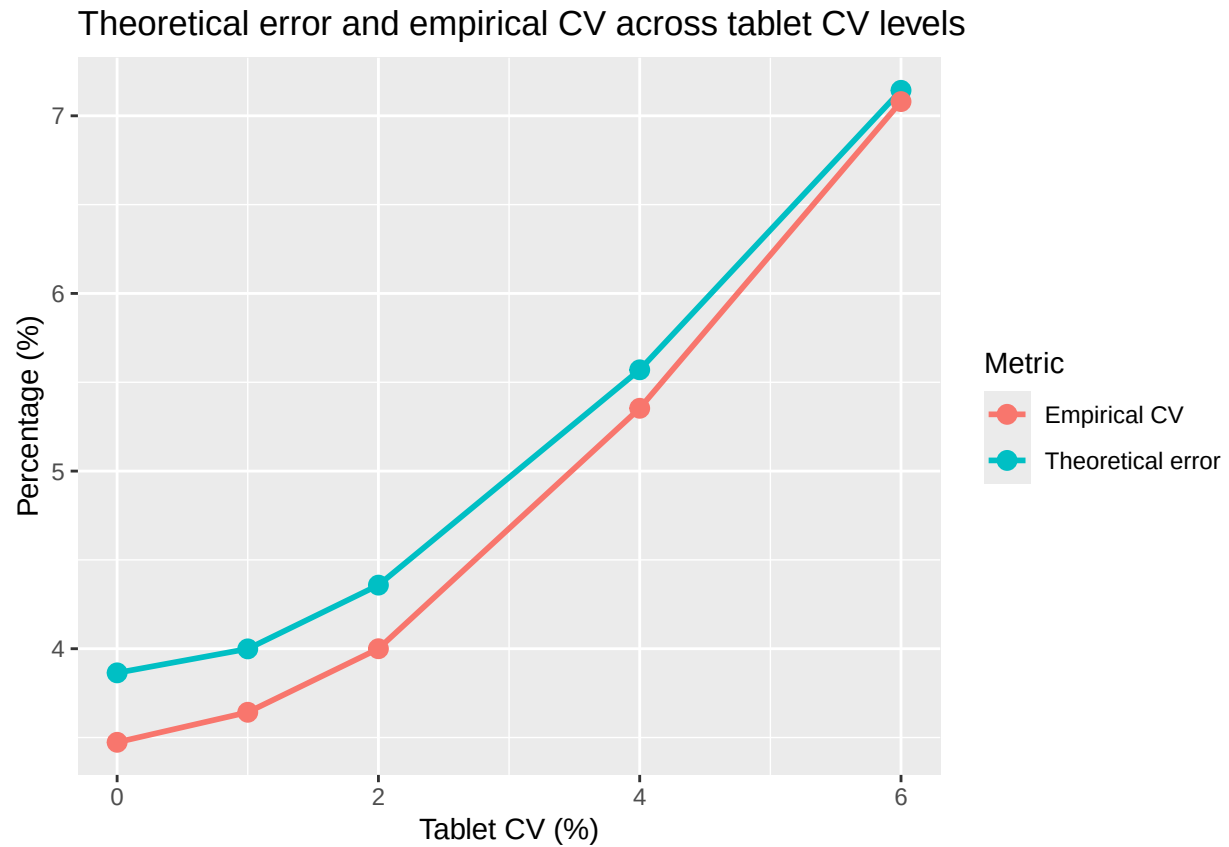
```

```
y = "Mean total error (%)"
)
```



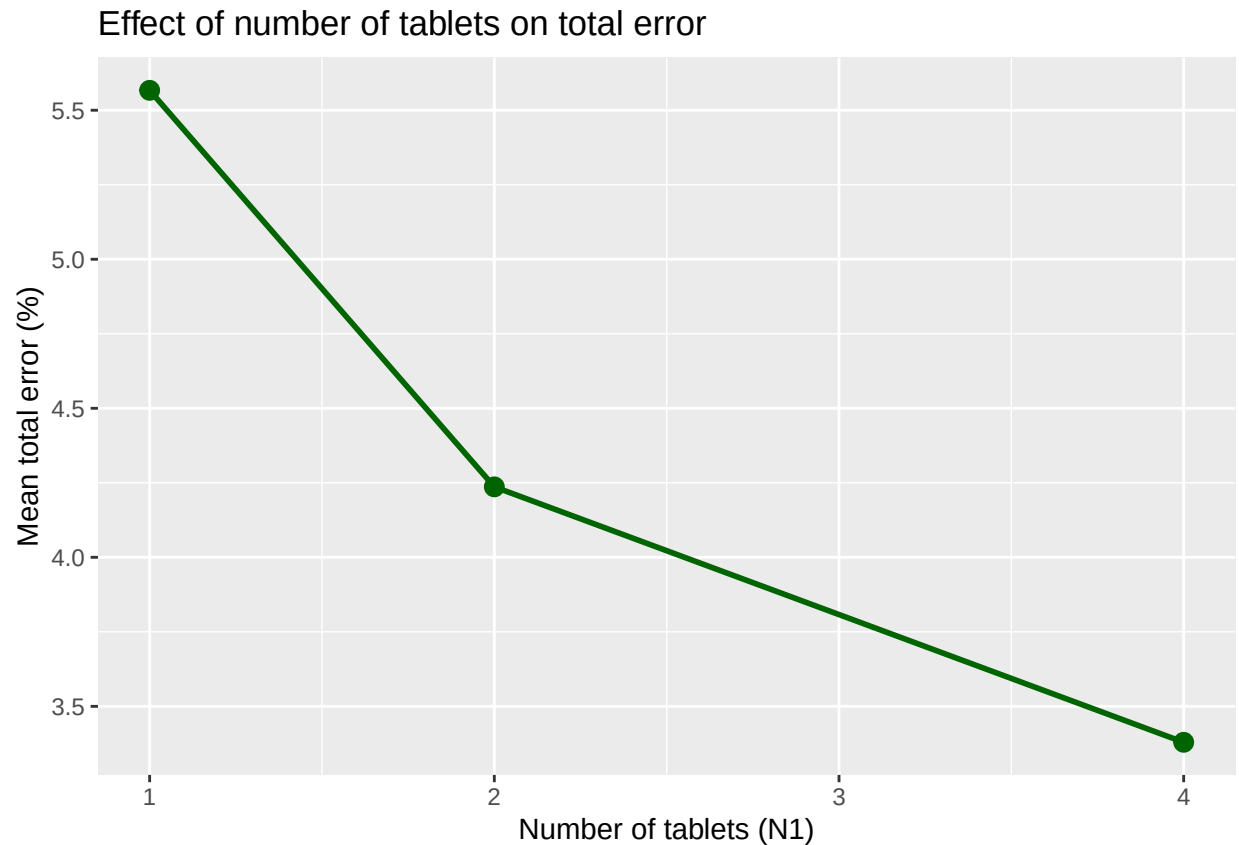
6 Plot 2: theoretical error and empirical CV

```
ggplot(summary_cv, aes(x = tablet_cv * 100)) +
  geom_line(aes(y = mean_theoretical_error, colour = "Theoretical error"), linewidth = 1) +
  geom_point(aes(y = mean_theoretical_error, colour = "Theoretical error"), size = 3) +
  geom_line(aes(y = empirical_cv, colour = "Empirical CV"), linewidth = 1) +
  geom_point(aes(y = empirical_cv, colour = "Empirical CV"), size = 3) +
  labs(
    title = "Theoretical error and empirical CV across tablet CV levels",
    x = "Tablet CV (%)",
    y = "Percentage (%)",
    colour = "Metric"
  )
)
```



7 Plot 3: number of tablets and total error

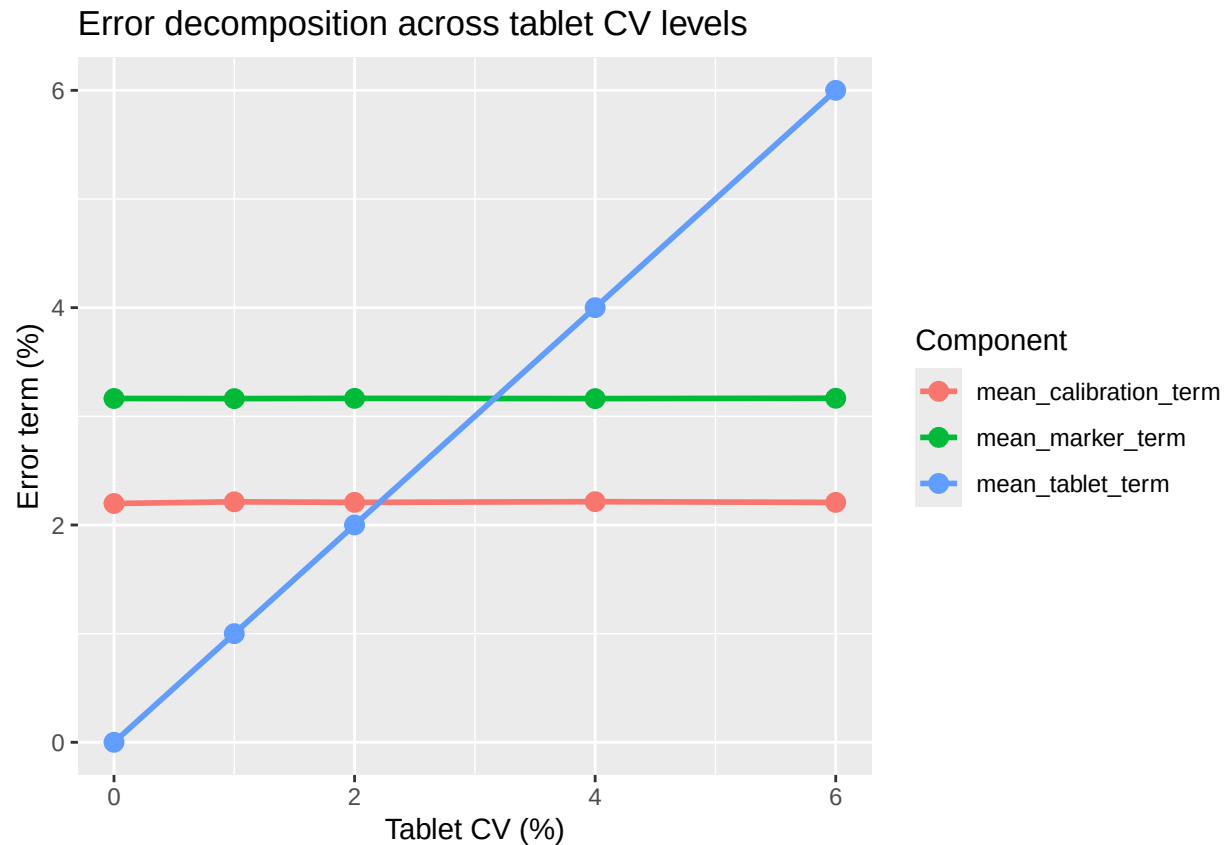
```
ggplot(summary_N1, aes(x = N1, y = mean_theoretical_error)) +
  geom_line(linewidth = 1, colour = "darkgreen") +
  geom_point(size = 3, colour = "darkgreen") +
  labs(
    title = "Effect of number of tablets on total error",
    x = "Number of tablets (N1)",
    y = "Mean total error (%)"
  )
```



Plot 4: tablet CV vs three error terms

```
summary_cv_long <- summary_cv %>%
  select(
    tablet_cv,
    mean_tablet_term,
    mean_calibration_term,
    mean_marker_term
  ) %>%
  tidyr::pivot_longer(
    cols = c(mean_tablet_term, mean_calibration_term, mean_marker_term),
    names_to = "term",
    values_to = "value"
  )

ggplot(summary_cv_long, aes(x = tablet_cv * 100, y = value, colour = term)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  labs(
    title = "Error decomposition across tablet CV levels",
    x = "Tablet CV (%)",
    y = "Error term (%)",
    colour = "Component"
  )
```

Plot 5: N1 vs three error terms

```
summary_N1_long <- summary_N1 %>%
  select(
    N1,
    mean_tablet_term,
    mean_calibration_term,
    mean_marker_term
  ) %>%
  tidyr::pivot_longer(
    cols = c(mean_tablet_term, mean_calibration_term, mean_marker_term),
    names_to = "term",
    values_to = "value"
  )

ggplot(summary_N1_long, aes(x = N1, y = value, colour = term)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  labs(
    title = "Error decomposition across numbers of tablets",
    x = "Number of tablets (N1)",
    y = "Error term (%)",
    colour = "Component"
  )
```

)

